Project Report: Optimization (AIT 203)

Question 1: Regression on Bike Sharing Demand

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1. Overview

Our goal for this question was to predict the hourly demand for rental bikes using the Kaggle "Bike Sharing Demand" dataset[1]. We tested different regression models to see how well they could learn patterns in data like temperature, humidity, and time of day. The main challenge was to balance "bias" (making the model too simple) and "variance" (making it too complex)[2].

2. Our Approach

Data Cleaning and Preprocessing

Before training, we had to clean the data to make sure our results were valid:

- **Avoiding "Cheating":** We noticed that the dataset includes columns for casual and registered users. Since casual + registered = count (our target), using these would let the model cheat and get a perfect score. We removed these columns immediately.
- **Feature Engineering:** The original data had a timestamp string. We extracted the **hour** and **month** from this, because bike rentals depend heavily on the time of day (e.g., morning rush hour) and the season.
- **Scaling:** We used a StandardScaler to bring all features (like windspeed and temperature) to the same scale so the model wouldn't be biased toward larger numbers.

Model Selection

We split the data into a training set (80%) and a testing set (20%)4. We then trained three types of models:

1. **Linear Regression:** A baseline model to see how a simple straight line performs.
2. **Polynomial Regression (Degrees 2, 3, 4):** We added powered terms (like temp^2 and humidity^3) to capture curves in the data, but we strictly turned off interaction terms for this part.
3. **Quadratic with Interaction:** We allowed variables to multiply with each other (e.g., temp \times humidity) to see if combined features helped prediction.

3. Results

Here are the metrics we obtained on the unseen test set:
|output:

```
Attempting to load 'train.csv'...
 -> Success! Loaded 10886 rows.

=== Question 1 Results: Bike Sharing Demand ===

Model: Linear Regression
  MSE: 21660.42
  R2 : 0.3438
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Model: Polynomial (d=2)
  MSE: 18301.86
  R2 : 0.4455
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Model: Polynomial (d=3)
  MSE: 16799.84
  R2 : 0.4910
----------------------------------------
Model: Polynomial (d=4)
  MSE: 16720.82
  R2 : 0.4934
----------------------------------------
Model: Quadratic (Interaction)
  MSE: 16557.63
  R2 : 0.4984
----------------------------------------
Generating graphs...
```
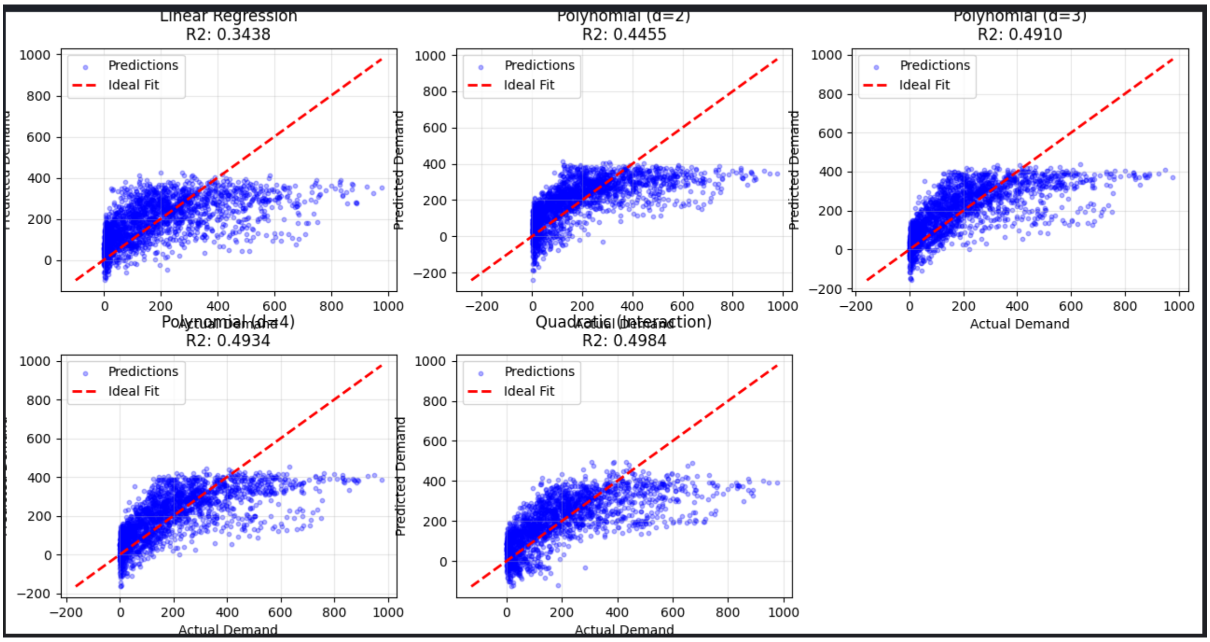

Model Type	**MSE (Lower is better)**	**R2 Score (Accuracy)**
Linear Regression (Baseline)	21,660.42	0.3438
Polynomial (Degree 2, No Interaction)	18,301.86	0.4455
Polynomial (Degree 3, No Interaction)	16,799.84	0.4910
Polynomial (Degree 4, No Interaction)	16,720.82	0.4934
Quadratic (With Interaction)	**16,557.63**	**0.4984**

Graphs:



4. Discussion and Interpretation

Why Linear Regression Failed (Underfitting)

Our baseline Linear model only achieved an R^2 of about 34%. This is a clear case of high bias (underfitting). Real-world behavior isn't linear—for example, demand doesn't just go up forever as the hour increases; it goes up in the morning, down at noon, and up again in the evening. A straight line simply couldn't capture these fluctuations.

Polynomials and Complexity

As we increased the complexity to Degree 2 and 3, the model performance improved significantly (jumping from 34% to 49%). This confirms that the relationship between weather and bike rentals is non-linear (curved). However, moving from Degree 3 to Degree 4 gave us almost no improvement (0.4910 vs 0.4934). This suggests that adding more complexity beyond this point might just lead to overfitting, where the model starts memorizing noise instead of learning patterns.

The Winner: Interaction Effects

The best performing model was the Quadratic Model with Interaction, which reached an (R^2) of nearly 50%.

This makes sense intuitively because environmental factors often work together. For instance, high humidity might not be bad on a cool day, but "High Humidity AND High Temperature" combined makes people less likely to rent bikes. The interaction terms allowed our model to learn these combined effects, which the standard polynomial models missed.

5. Conclusion

Based on our test metrics, the **Quadratic Interaction Model** is the best choice. It provided the lowest error (MSE: 16,557) and effectively captured the complex, non-linear relationships in the data without becoming unnecessarily complicated.